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Course code: 6BUIS017W1

Customer Relationship & Change Management Coursework 2



Task A

The complete Python code for Task A was written and run using Google Colab. Due to the code's extensive length and complexity, it is impractical to be presented in full within the report; therefore links to the complete notebooks are provided. The key output values generated after the execution of the Python code are discussed in the report, along with vital figures.

Questions A-F:

<https://colab.research.google.com/drive/1aBxddDD86nZKcouY0r1pO8diwKchvpWR?usp=sharing>

Question G:

https://colab.research.google.com/drive/1roWNOE4o1mOdcHIOZbuDDXz82q_KWfUb?usp=sharing

Task A Outputs and Explanation

a) Amazon Closing Stock Price chart last 5 years



This line chart displays the closing stock price of Amazon over the last five years, both in the short and long term. From 2021 through to early 2023, the stock has seen significant volatility, including a steep drop at the end of 2022 and the beginning of 2023 - a probable consequence of wider market corrections amidst economic uncertainties. The price follows an unmistakable upward trajectory from mid-2023 onward, recovering steadily to mark new highs by 2025, though it still sees periodic setbacks. The following visualisation underlines the cyclic nature of this stock, the growth trajectory that the stock is taking, and the related volatility that predictive models need to consider for future price estimations.

b) Calculating technical indicators for stock prediction

	Open	High	Low	Close	Volume	Dividends	Stock Splits	SMA_20	SMA_50	EMA_20	RSI
Date											
2021-02-19 00:00:00-05:00	166.411499	166.675003	162.287506	162.494995	86104000	0.0	0.0	164.793499	161.549140	164.013116	54.100509
2021-02-22 00:00:00-05:00	160.406494	161.615997	158.613007	159.037003	70314000	0.0	0.0	164.514774	161.552590	163.539200	32.578340
2021-02-23 00:00:00-05:00	156.351501	160.236496	154.679993	159.725006	93544000	0.0	0.0	164.266025	161.642890	163.175944	29.014900
2021-02-24 00:00:00-05:00	158.337494	158.561493	156.268997	157.976501	60226000	0.0	0.0	163.849525	161.700930	162.680759	31.317783
2021-02-25 00:00:00-05:00	156.837006	158.912994	152.388000	152.858002	90676000	0.0	0.0	163.410975	161.641670	161.745258	22.248558
...	...	...	...	...	...	...	...	...	...	...	...
2025-12-02 00:00:00-05:00	235.009995	238.970001	233.550003	234.419998	45785400	0.0	0.0	235.175500	227.751201	231.453861	35.476840
2025-12-03 00:00:00-05:00	233.350006	233.380005	230.610001	232.380005	35495100	0.0	0.0	234.328500	227.984601	231.542065	37.604868
2025-12-04 00:00:00-05:00	232.770004	233.500000	226.800003	229.110001	45683200	0.0	0.0	233.274001	228.162600	231.310440	40.446646
2025-12-05 00:00:00-05:00	230.320007	231.240005	228.550003	229.529999	33075500	0.0	0.0	232.598501	228.390201	231.140874	43.836592
2025-12-08 00:00:00-05:00	229.520004	230.824005	227.080002	227.194397	16189725	0.0	0.0	231.737720	228.538488	230.765019	43.303221

1207 rows × 11 columns

	EMA_20	SMA_50
Date		
2021-01-04 00:00:00-05:00	159.331497	NaN
2021-01-05 00:00:00-05:00	159.483308	NaN
2021-01-06 00:00:00-05:00	159.239088	NaN
2021-01-07 00:00:00-05:00	159.131366	NaN
2021-01-08 00:00:00-05:00	159.131711	NaN

close

	Open	High	Low	Close	Volume	Dividends	Stock Splits	SMA_20	SMA_50	EMA_20	RSI
Date											
2021-03-16 00:00:00-04:00	155.248505	156.445496	153.792999	154.593002	50776000	0.0	0.0	155.610976	159.487180	155.459210	44.758087
2021-03-17 00:00:00-04:00	153.660995	158.652496	153.511002	156.786499	62372000	0.0	0.0	155.178701	159.436280	155.585619	56.692839
2021-03-18 00:00:00-04:00	155.050003	155.831497	151.250000	151.399506	72992000	0.0	0.0	154.428101	159.245760	155.186941	45.072399
2021-03-19 00:00:00-04:00	151.461502	153.864502	150.831497	153.748001	92508000	0.0	0.0	153.990752	159.182340	155.049899	44.547255
2021-03-22 00:00:00-04:00	153.392502	156.328995	153.002502	155.543503	58044000	0.0	0.0	153.816077	159.131050	155.096909	51.282577
...	...	...	...	...	...	...	...	...	...	...	...
2025-12-24 00:00:00-05:00	232.130005	232.949997	231.330002	232.380005	11420500	0.0	0.0	228.909500	230.069200	228.715411	55.521790
2025-12-26 00:00:00-05:00	232.039993	232.990005	231.179993	232.520004	15994700	0.0	0.0	229.077500	230.408200	229.077753	55.097179
2025-12-29 00:00:00-05:00	231.940002	232.600006	230.770004	232.070007	19797900	0.0	0.0	229.020000	230.760200	229.362729	59.543125
2025-12-30 00:00:00-05:00	231.210007	232.770004	230.199997	232.529999	21910500	0.0	0.0	228.952500	231.150000	229.664374	58.675201
2025-12-31 00:00:00-05:00	232.910004	232.990005	230.119995	230.820007	24358700	0.0	0.0	228.772501	231.436801	229.774434	48.034414

1206 rows × 11 columns

	Date
2021-01-04 00:00:00-05:00	159.331497
2021-01-05 00:00:00-05:00	160.925507
2021-01-06 00:00:00-05:00	156.919006
2021-01-07 00:00:00-05:00	158.108002
2021-01-08 00:00:00-05:00	159.134995

dtype: float64

Four technical variables have been used to forecast the stock price of Amazon. These variables include SMA_20, EMA_20, SMA_50, and RSI. The variable SMA_20 represents the 20-day Simple Moving Average. This attribute charts the short-term movements of the stock price. EMA_20, which shows the 20-day Exponential Moving Average, assigns more weight to the latest stock price and can therefore quickly reflect changes. The SMA_50 attribute represents the 50-day Simple Moving Average and reflects the stock price's long-term movement. The Relative Strength Index, measured over a 14-day cycle, indicates stock price momentum. When the index is above 70, it reveals overbuying, while a position below 30 reveals overselling. The starting observations that contain NaN values, which result from the rolling window, have been eliminated.

c) Train and Test Data

In this step, the data is split into a training and testing data set so that the model is able to recognise patterns in past data and make forecasts on unseen data. 70% of the data available is used for training of the model, and 30% is held back and used for testing. This data division is done in a sequential manner and not randomly, meaning that data that comes before is used for training and data that comes later is used for testing. Doing so ensures that data from before the test point in a time series, like stock prices, is used for training and data after that is used for testing. Both $X[:t]$ and $y[:t]$ refer to data from before the split point for the training of features and target variables respectively, while $X[t:]$ and $y[t:]$ refer to data after the split point for testing.

	EMA_20	SMA_50
Date		
2021-03-19 00:00:00-04:00	155.042310	159.182340
2021-03-22 00:00:00-04:00	155.090043	159.131050
2021-03-23 00:00:00-04:00	155.260039	159.085850
2021-03-24 00:00:00-04:00	155.173702	159.058710
2021-03-25 00:00:00-04:00	154.901254	158.984140
...	...	...
2024-07-22 00:00:00-04:00	190.337445	187.558800
2024-07-23 00:00:00-04:00	189.963403	187.497000
2024-07-24 00:00:00-04:00	189.093555	187.364001
2024-07-25 00:00:00-04:00	188.213217	187.229601
2024-07-26 00:00:00-04:00	187.669101	187.138200

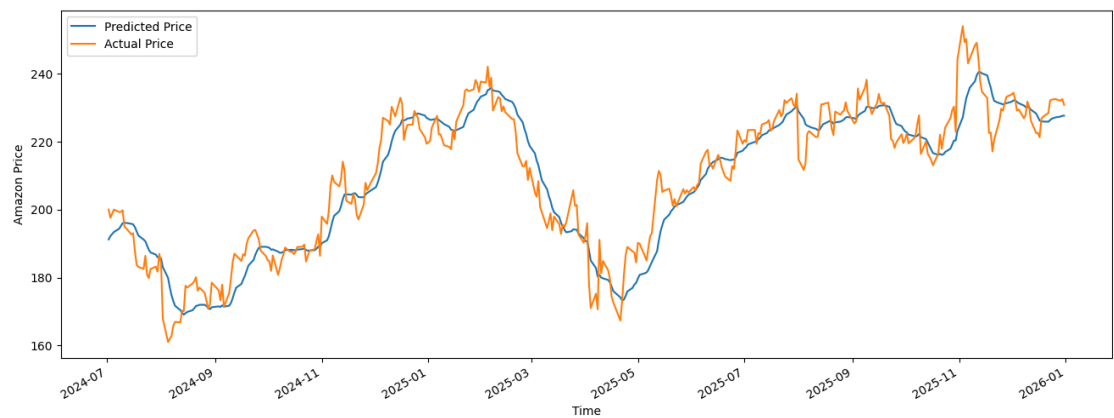
844 rows x 2 columns

d) Build and Train the Linear Regression Model

... Linear Regression Model Defined and Trained

In this part of the task, the dataset is divided into training and testing parts to train the Linear Regression Model using past observations and test the results based on unseen values in the dataset. A split of 70 and 30 is used, where the first 70% of the dataset is used for training, and the remaining 30% for evaluating its predictions, which is critical in time-series analysis to ensure the results are based solely on the available data at the point of prediction. Slices $X[:t]$ and $y[:t]$ are used for training the model, while $X[t:]$ and $y[t:]$ are used for testing to ensure the accuracy of the results obtained by predicting the values based on the moving averages of EMA_20 and SMA_50 as the input used to predict the future stock values.

e) Prediction Process and Results



The model has a 89.60% accuracy.

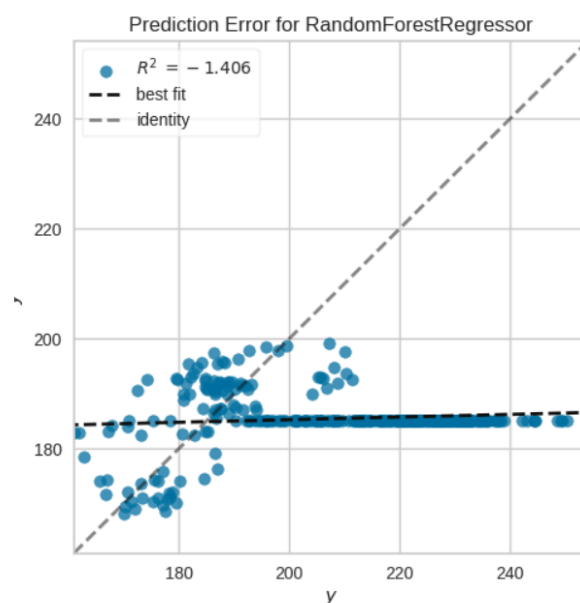
During this step, the linear regression model that was trained was used on the test dataset to predict stock prices for Amazon. Then, these predictions were plotted against the actual observed values over the same time period to visually understand the performance of this model. From the graph, it is seen that the predicted and actual prices are well aligned with some slight deviations due to market volatility. To quantify the accuracy of this model, the R-squared score was computed, which returned an accuracy value of 89.60%, a pretty strong accuracy showing that this model explains a significant proportion of variance in stock prices and reliably performs well on unseen data.

f) Alpha, Beta Values and Linear Regression Equation

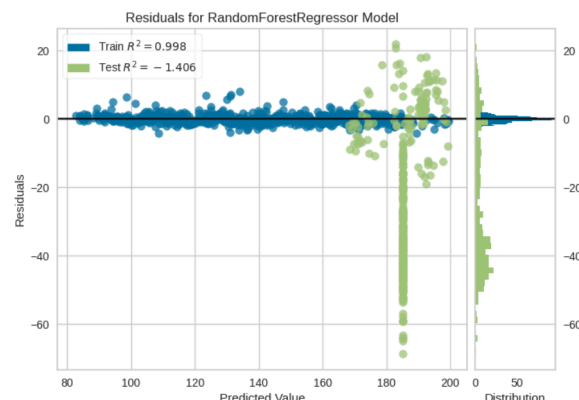
```
alpha = 2.7299279775985212
betas = [ 1.51557027 -0.53406647]
Predicted Price = 2.7299279775985212 + (1.515570269549142 * EMA_20) + (-0.5340664681777866 * SMA_50)
```

During this step, the alpha and beta values derived from the linear regression model were obtained using Python. It was observed that the value of alpha was **2.7299**, whereas the values related to **EMA_20** and **SMA_50** were **1.5156** and **-0.5341**, respectively. Based on these values, the formula for the linear regression model could be established, which was represented as follows: **Predicted Price = 2.7299 + (1.5156 × EMA_20) + (- 0.5341 × SMA_50)**. This equation indicates how each moving average makes a contribution to the prediction, and it provides a highly accurate representation of the model's dynamics.

g) Random Forest vs. Linear Regression: Implementation and Evaluation



Next, the Random Forest Regressor was built taking MA20, EMA20, MA50, and RSI as input variables to predict Amazon stock movement and compared to the Linear Regression model. The method was trained and tested based on the historical records, where the model performance was measured using the prediction error and residual plots, along with a comparative plot of the actual and predicted stock prices over a timeframe. The plots show that while the Random Forest model fits the training data extremely well, it exhibits weaker predictive accuracy when it comes to new input examples, which raises concerns about overfitting. The Linear Regression model, being more basic, showcases more consistent performance over the test phase. This example illustrates that while complex models using ensemble techniques may successfully identify complicated patterns, simpler models may actually be more successful at financial time series predictions in volatile datasets. The slightly worse performance on out-of-sample data as well as the lower value of R^2 achieved by the Random Forest model could be accounted for by less optimisation of hyperparameters as well as the high noise level that generally characterises financial time series.



Task B

How the Financial Technologies Industry is Implementing Data-Driven Change Management Strategies

Abstract

The rapid growth of Financial Technologies (FinTech) has fundamentally reshaped how financial services operate, compete, and grow. A key part of this change is the use of data-driven and predictive modelling approaches that enables organisations to anticipate change rather than responding to it. This paper explores how the FinTech industry implements data-driven change management strategies, focusing on the implications of intelligent finance for organisational change. It argues that predictive analytics not only improves decision-making but also affects leadership, company culture, and operations. The paper concludes by outlining key considerations for organisations seeking to manage change effectively in an increasingly data-centric financial environment.

Introduction

The finance sector has traditionally been conservative, highly regulated, and slow to change. However, the emergence of FinTech companies has introduced a new dynamic by leveraging sophisticated data analytics, machine learning, and artificial intelligence. These technologies help firms to process large volumes of financial data and respond rapidly to market and

operational demands. As a result, FinTech firms increasingly rely on data-driven change management, making it essential to understand how intelligent finance influences company transformation.

Data-Driven Change Management in FinTech

Change management involves the process of transitioning individuals, teams, and processes from the current state to the future desired state. In the FinTech sector, evidence-based change management replaces the need for intuition-driven decision making. This is because the industry relies on predictive analytics to determine the risk associated with the future desired change before implementation.

For instance, these firms rely on past data of transactions and behavior patterns of consumers to determine the demand for new online products. The data helps in deciding the need for system upgrades, training, and customer engagement. By utilising data for change, the organisations face less uncertainty, and confidence among the stakeholders contributes to the success of change adoption.

Implications of Intelligent Finance on Change Management

Intelligent finance refers to the integration of AI, machine learning, and advanced analytics into financial processes. Its impact on change management is significant. First, intelligent finance enables real-time monitoring of organisational performance, allowing leaders to adjust change strategies dynamically rather than following rigid implementation plans. This adaptive approach is particularly valuable in volatile markets where conditions can change rapidly.

Second, intelligent finance reshapes organisational culture. Employees are increasingly expected to trust data-driven insights, which requires a shift in mindset and skillsets. Change management strategies must therefore address resistance to algorithmic decision-making and promote data literacy across FinTech firms. Without this cultural alignment, even the most advanced predictive models may fail to deliver value.

Lastly, intelligent finance introduces ethical and governance challenges. Predictive models can reinforce bias or produce unclear decisions, which may undermine trust among employees and customers. Effective change management must incorporate transparency, explainability, and compliance considerations to ensure that data-driven changes align with regulatory and ethical standards.

Predictive Modelling as a Driver of Organisational Change

Predictive modelling plays a central role in enabling proactive change management within FinTech organisations. By forecasting customer churn, credit risk, fraud likelihood, or market volatility, predictive models allow firms to design targeted interventions before problems escalate. This proactive capability contrasts with traditional reactive approaches, where change occurs only after negative outcomes are observed. For instance, Monzo employs machine learning fraud detection tools, helping it identify unusual spending habits in real-time, illustrating

how predictive analytics can directly influence changes in operations and enhance consumer protection.

From a change management perspective, predictive modelling supports scenario planning and risk mitigation. Leaders can simulate the effects of strategic decisions, such as launching a new digital platform or automating back-office operations, and assess their potential impact on performance and employee roles. This reduces implementation risk and enhances decision quality.

Managing Change Effectively in a Data-Driven Environment

To implement change successfully in a data-driven FinTech ecosystem, organisations must align technology, people, and processes. Moreover, leadership commitment is essential since leaders must support data-driven decision making. Training and development programs also play a key role in enabling employees to work with predictive insights.

Companies should implement change through a phased approach instead of organisational transformation. Using predictive analytics, firms can measure the effect of small-scale changes. Transparency in communication should also be a goal. Stakeholders need to be aware not only of the organisational changes taking place, but also how data can help create these changes.

Conclusion

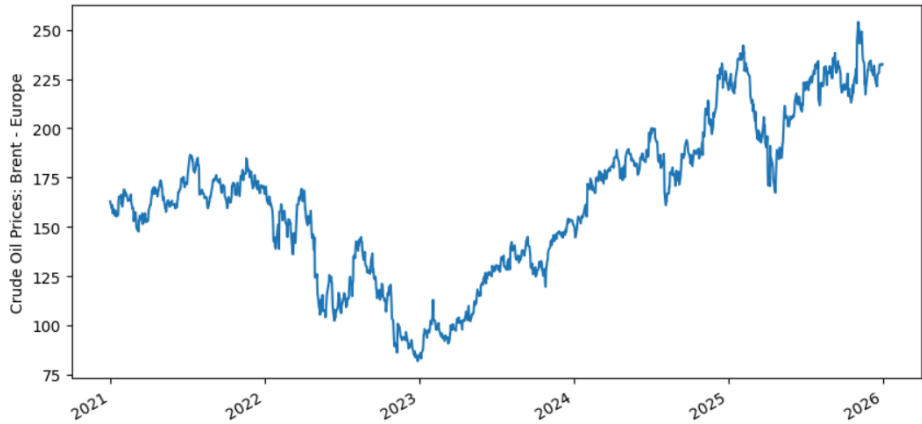
The FinTech industry demonstrates how data-driven and predictive modelling approaches can fundamentally reshape change management strategies. Intelligent finance enables organisations to anticipate challenges, adapt to market conditions, and carry out changes effectively. However, the success of these approaches depends on cultural readiness, ethical practices, and effective leadership. Companies that integrate predictive analytics into their change management frameworks are better positioned to achieve sustainable development and long-term competitive advantage in today's increasingly complex financial market.

References

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3. Blackboard: Week 8 tutorial
4. [A Predictive Model of the Stock Market Using the LSTM Algorithm with a Combination of Exponential Moving Average \(EMA\) and Relative Strength Index \(RSI\) Indicators - Astrophysics Data System](https://www.astronexus.com/papers/2025/4/51860.pdf)
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6. <https://www.nature.com/articles/s41599-023-01923-4>
7. <https://eelet.org.uk/index.php/journal/article/view/1762>

Appendix

<Axes: xlabel='Date', ylabel='Crude Oil Prices: Brent - Europe'>



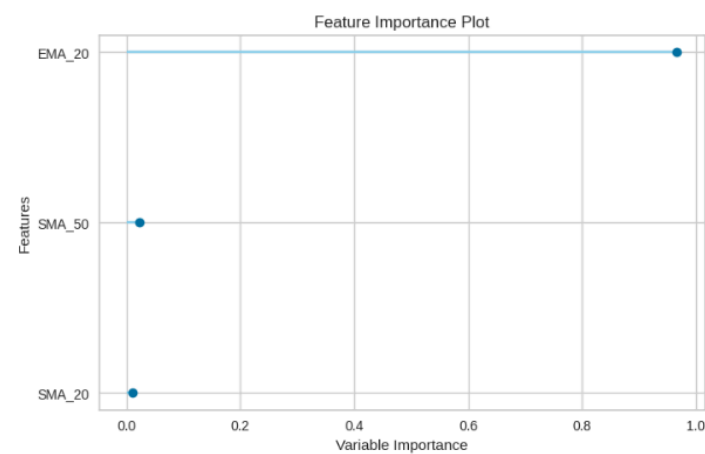
	MAE	MSE	RMSE	R2	RMSLE	MAPE
Fold						
0	2.5346	10.6179	3.2585	0.7337	0.0187	0.0149
1	2.9109	13.1875	3.6315	0.8925	0.0230	0.0181
2	4.3146	28.0398	5.2953	0.9309	0.0380	0.0317
3	3.7977	22.1654	4.7080	0.8449	0.0395	0.0317
4	3.0134	15.2110	3.9001	0.8976	0.0392	0.0307
5	1.3668	3.0295	1.7405	0.8421	0.0169	0.0136
6	1.7767	5.1321	2.2654	0.9309	0.0174	0.0139
7	2.6879	10.8336	3.2915	0.8163	0.0247	0.0201
8	2.0975	6.5536	2.5600	0.9494	0.0158	0.0129
9	2.7071	10.6437	3.2625	0.7337	0.0175	0.0146
Mean	2.7207	12.5414	3.3913	0.8572	0.0251	0.0202
Std	0.8349	7.2925	1.0200	0.0740	0.0094	0.0076

	MAE	MSE	RMSE	R2	RMSLE	MAPE
Fold						
0	1.9327	6.5512	2.5595	0.8357	0.0148	0.0113
1	2.9272	13.4004	3.6606	0.8908	0.0232	0.0182
2	3.9909	24.2538	4.9248	0.9402	0.0348	0.0287
3	3.7040	21.5529	4.6425	0.8492	0.0390	0.0310
4	3.0085	15.1775	3.8958	0.8979	0.0392	0.0306
5	1.3659	3.0503	1.7465	0.8410	0.0170	0.0136
6	1.8057	5.2573	2.2929	0.9292	0.0177	0.0141
7	2.7035	10.9025	3.3019	0.8151	0.0247	0.0202
8	2.0588	6.4086	2.5315	0.9505	0.0156	0.0127
9	2.7154	10.6925	3.2699	0.7324	0.0175	0.0146
Mean	2.6213	11.7247	3.2826	0.8682	0.0243	0.0195
Std	0.7962	6.6494	0.9743	0.0636	0.0093	0.0074

	Description	Value
0	Session id	123
1	Target	Close
2	Target type	Regression
3	Original data shape	(1205, 4)
4	Transformed data shape	(1205, 4)
5	Transformed train set shape	(843, 4)
6	Transformed test set shape	(362, 4)
7	Numeric features	3
8	Preprocess	True
9	Imputation type	simple
10	Numeric imputation	mean
11	Categorical imputation	mode
12	Fold Generator	TimeSeriesSplit
13	Fold Number	10
14	CPU Jobs	-1
15	Use GPU	False
16	Log Experiment	False
17	Experiment Name	reg-default-name
18	USI	9idf

ID	Name	Reference	Turbo
lr	Linear Regression	sklearn.linear_model_base.LinearRegression	True
lasso	Lasso Regression	sklearn.linear_model_coordinate_descent.Lasso	True
ridge	Ridge Regression	sklearn.linear_model_ridge.Ridge	True
en	Elastic Net	sklearn.linear_model_coordinate_descent.Elast...	True
lar	Least Angle Regression	sklearn.linear_model_least_angle.Lars	True
llar	Lasso Least Angle Regression	sklearn.linear_model_least_angle.LassoLars	True
omp	Orthogonal Matching Pursuit	sklearn.linear_model_omp.OrthogonalMatchingPu...	True
br	Bayesian Ridge	sklearn.linear_model_bayes.BayesianRidge	True
ard	Automatic Relevance Determination	sklearn.linear_model_bayes.ARDRRegression	False
par	Passive Aggressive Regressor	sklearn.linear_model_passive_aggressive.Passi...	True
ransac	Random Sample Consensus	sklearn.linear_model_ransac.RANSACRegressor	False
tr	TheilSen Regressor	sklearn.linear_model_theil_sen.TheilSenRegressor	False
huber	Huber Regressor	sklearn.linear_model_huber.HuberRegressor	True
kr	Kernel Ridge	sklearn.kernel_ridge.KernelRidge	False
svm	Support Vector Regression	sklearn.svm_classes.SVR	False
knn	K Neighbors Regressor	sklearn.neighbors_regression.KNeighborsRegressor	True
dt	Decision Tree Regressor	sklearn.tree_classes.DecisionTreeRegressor	True
rf	Random Forest Regressor	sklearn.ensemble_forest.RandomForestRegressor	True
et	Extra Trees Regressor	sklearn.ensemble_forest.ExtraTreesRegressor	True
ada	AdaBoost Regressor	sklearn.ensemble_weight_boosting.AdaBoostRegr...	True
gbr	Gradient Boosting Regressor	sklearn.ensemble_gb.GradientBoostingRegressor	True
mlp	MLP Regressor	sklearn.neural_network_multilayer_perceptron...	False
xgboost	Extreme Gradient Boosting	xgboost.sklearn.XGBRegressor	True
lightgbm	Light Gradient Boosting Machine	lightgbm.sklearn.LGBMRegressor	True
dummy	Dummy Regressor	sklearn.dummy.DummyRegressor	True

	MAE	MSE	RMSE	R2	RMSLE	MAPE
old						
0	10.6139	143.3746	11.9739	-2.5953	0.0688	0.0636
1	4.8793	54.8704	7.4075	0.5529	0.0473	0.0313
2	21.5927	780.8202	27.9432	-0.9238	0.2044	0.1781
3	12.6414	216.1081	14.7006	-0.5117	0.1175	0.0997
4	15.6981	327.7746	18.1045	-1.2056	0.1791	0.1718
5	8.6051	91.9345	9.5882	-3.7923	0.0981	0.0849
6	7.3795	88.7154	9.4189	-0.1941	0.0775	0.0583
7	7.5518	79.7896	8.9325	-0.3529	0.0657	0.0542
8	4.4567	26.4635	5.1443	0.7957	0.0308	0.0269
9	12.6981	200.2737	14.1518	-4.0113	0.0773	0.0673
ean	10.6117	201.0124	12.7365	-1.2238	0.0966	0.0836
itd	4.9951	211.3823	6.2284	1.6093	0.0530	0.0501



	MAE	MSE	RMSE	R2	RMSLE	MAPE
Fold						
0	9.5493	123.2058	11.0998	-2.0895	0.0640	0.0573
1	4.9521	53.5902	7.3205	0.5633	0.0468	0.0317
2	20.8654	748.0342	27.3502	-0.8430	0.2008	0.1727
3	12.5411	219.1156	14.8026	-0.5328	0.1182	0.0983
4	16.6731	375.3491	19.3739	-1.5257	0.1903	0.1831
5	8.5444	88.8467	9.4259	-3.6313	0.0963	0.0843
6	6.3362	70.8633	8.4180	0.0462	0.0695	0.0501
7	7.4785	79.6693	8.9258	-0.3509	0.0651	0.0534
8	3.5788	17.9043	4.2313	0.8618	0.0254	0.0216
9	15.4129	279.0733	16.7055	-5.9830	0.0922	0.0819
Mean	10.5932	205.5652	12.7654	-1.3485	0.0969	0.0834
Std	5.3258	210.0952	6.5277	1.9918	0.0552	0.0523



	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE
0	Random Forest Regressor	27.1746	1037.6931	32.2132	-1.4058	0.1561	0.1228
Date							
2024-07-23 00:00:00-04:00		193.123505	187.558807	189.963409	186.410004		190.943401
2024-07-24 00:00:00-04:00		193.165497	187.496994	189.093552	180.830002		188.869900
2024-07-25 00:00:00-04:00		192.889999	187.363998	188.213211	179.850006		192.637101
2024-07-26 00:00:00-04:00		192.201996	187.229599	187.669098	182.500000		193.731402
2024-07-29 00:00:00-04:00		191.434494	187.138199	187.243469	183.199997		194.788302
...		...	...	...	...		...
2025-12-23 00:00:00-05:00		228.481003	229.507996	228.329666	232.139999		185.192799
2025-12-24 00:00:00-05:00		228.774002	229.749405	228.715408	232.380005		185.192799
2025-12-26 00:00:00-05:00		228.909500	230.069199	229.077759	232.520004		185.192799
2025-12-29 00:00:00-05:00		229.077499	230.408203	229.362732	232.070007		185.192799
2025-12-30 00:00:00-05:00		229.020004	230.760193	229.664368	232.529999		185.192799

362 rows × 5 columns

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE
0	Random Forest Regressor	1.2256	3.0332	1.7416	0.9930	0.0085	0.0059
Date							
2024-07-23 00:00:00-04:00		193.123505	187.558807	189.963409	186.410004		185.764003
2024-07-24 00:00:00-04:00		193.165497	187.496994	189.093552	180.830002		182.685502
2024-07-25 00:00:00-04:00		192.889999	187.363998	188.213211	179.850006		181.416104
2024-07-26 00:00:00-04:00		192.201996	187.229599	187.669098	182.500000		182.327400
2024-07-29 00:00:00-04:00		191.434494	187.138199	187.243469	183.199997		183.026199
...		...	...	...	...		...
2025-12-23 00:00:00-05:00		228.481003	229.507996	228.329666	232.139999		230.232400
2025-12-24 00:00:00-05:00		228.774002	229.749405	228.715408	232.380005		231.755204
2025-12-26 00:00:00-05:00		228.909500	230.069199	229.077759	232.520004		232.045904
2025-12-29 00:00:00-05:00		229.077499	230.408203	229.362732	232.070007		231.988006
2025-12-30 00:00:00-05:00		229.020004	230.760193	229.664368	232.529999		232.238802

362 rows × 5 columns